48V Electrical Architecture

Document role

- This file owns the final `48V` electrical architecture for the active build baseline.
- Use it for house-bank structure, alternator-path direction, shutdown logic, and manual control-path intent.
- Keep full conductor schedules, fuse matrices, and layout-level implementation detail in `docs/implementation/`.
- Keep unresolved vendor gates, risk state, and follow-up closure items in `docs/core/TRACKING.md`.
- Keep broad project sequencing and day-to-day execution framing in `docs/core/PROJECT.md` or the active plan docs.

As-of date: `2026-07-19`

Purpose: hold the finalized, concise `48V` house and alternator architecture in one place so wiring, protection, shutdown behavior, and BOM references are easy to understand without re-reading the historical trade studies.

Related docs:

- Electrical topology diagram
- Electrical fuse schedule
- Mechman / WS500 / APM-48 install guide
- Systems
- Tracking
- Estimated BOM

Final direction

- House architecture stays `48V` core with `3x 51.2V 100Ah` batteries in parallel.
- Alternator charging is the dedicated `48V` secondary alternator path; obsolete pre-Mechman charger hardware is removed from active planning.
- Protection baseline is `Mechman + WS500 + Balmar APM-48`.
- Manual alternator-charge enable/disable is through Ford `Upfitter Switch #3`.
- `Upfitter #3` is a low-current control signal only. It does not carry alternator output current.
- `WS500` white `Feature-In` is reserved for future fault-interlock work, not required in Phase 1.

Current commissioning state (`2026-07-19`)

- `48V` bus has been live-tested: owner measured `55.5V` throughout the system, including at the MultiPlus.
- MultiPlus-II DC/inverter mode has been switched on with inverter light illuminated, slight normal hum, and no reported error lights.
- SmartShunt and Orion-Tr Smart are visible in VictronConnect.
- Cerbo GX access point/remote-console workflow is active; Cerbo power is a small inline fused feed from the `48V` system side and MultiPlus communication is via `VE.Bus` RJ45.
- Shore charging has been tested through the MultiPlus at household-outlet current limits. The first short test proved basic AC-in/charger function; later owner verification redid the settings and confirmed first-battery behavior entered bulk, then quickly transitioned to absorption at/near `100%` as planned.
- MultiPlus LiFePO4 charge-profile programming/verification is treated as closed for the current shore-charger setup. Current target is `56.8V` absorption/charge, `54.0V` float, short absorption dwell, equalization off, conservative LiFePO4 storage behavior, and source-current limit matched to the actual shore circuit. The Dumfume manual's `58.4V +/-0.2V` value is documented, but because the same manual also lists `58.4V` as charge-limit/over-charge protection voltage, it is not the active routine charger target. `DVCC` remains disabled unless a documented BMS/GX control path is added.
- The electrical module is now hard-mounted through the finished floor to registered truck-bed hardpoints. The MultiPlus and AC breaker enclosure were removed for the lift and are pending easy front-side remount into embedded pronged T-nuts; final Bench tie-in still supplies the anti-rack/road-restraint path.
- All three batteries' `2/0 AWG` branch leads are cut, lugged, heat-shrunk, and landed at the battery-side positive/negative busbars. The three batteries are not yet paralleled: Batteries 2 and 3 remain on the individual-charge/rest/ `<=0.1V` matching gate.
- Permanent shore inlet/AC-in, full parallel-bank commissioning, `12V` branch closeout, AC-out branch/GFCI commissioning, secondary-alternator commissioning, final Cerbo mounting, and board strain-relief/abrasion-control remain separate future gates.

Locked component set

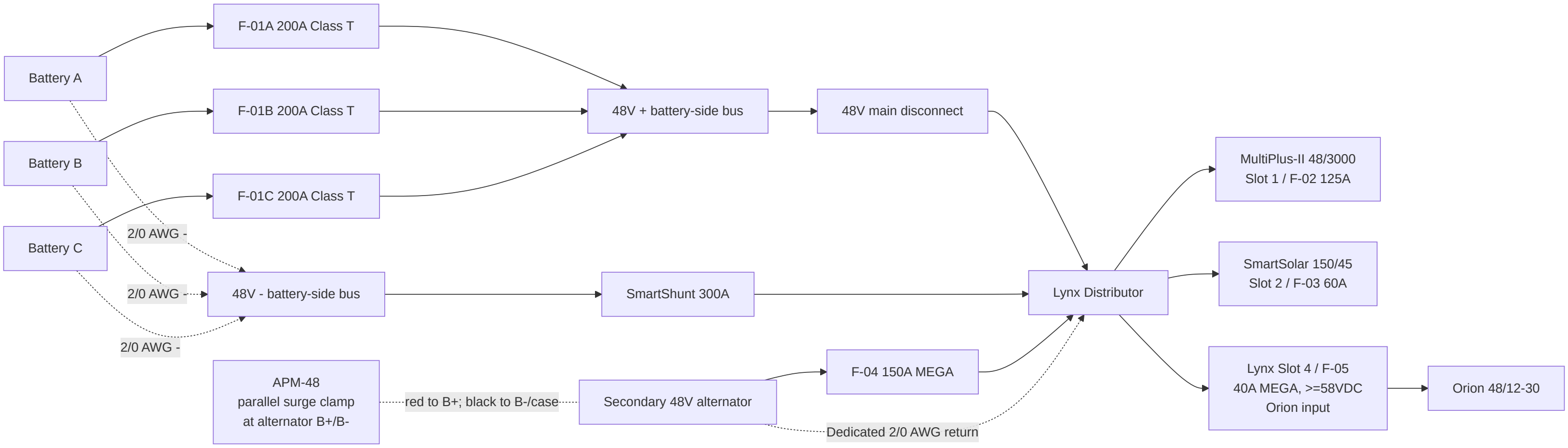
Function	Locked baseline	BOM row(s)

House batteries	3x Dumfume 51.2V 100Ah LiFePO4	3
Main house disconnect	Victron 275A battery switch	5
Main fused distribution	Victron Lynx Distributor M10	6
Battery branch protection	F-01A/B/C 200A Class T; manual-backed battery limit is 200A max continuous discharge per battery, with terminal/manufacturer fuse guidance still not separately specified; owner confirmed 3 holders and 4 slow-blow Class T fuses total	7
Inverter/charger	MultiPlus-II 48/3000/35-50	12
48V to 12V charger	Orion-Tr Smart 48/12-30; fed directly from Lynx Slot 4 through one verified 40A MEGA (58VDC minimum under the locked 56.8V charge ceiling; Victron CIP138040020 40A/80V is the replacement fallback) into the existing 6 AWG input pair; no separate inline/DIN input fuse; keep 6 AWG and F-07 60A/80V on 12V output	20, 10, 11
Monitoring	Cerbo GX + SmartShunt 300A	22, 23
Alternator kit	Mechman 48V secondary alternator kit with WS500	168
Load-dump clamp	Balmar APM-48	169
Alternator branch fuse	F-04 150A MEGA (58V/80V) in Lynx Slot 3	170
WS500 low-current fuse set	F-12 regulator power + F-13 positive voltage sense; current-sense pair is unfused per current manual	171
WS500 Upfitter #3 control kit	F-15 + control wire + holder/terminals	176

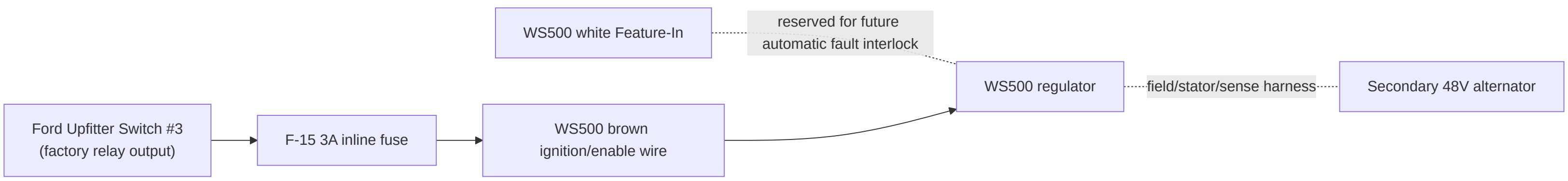
Battery manual limits and charger-programming baseline

- Battery model basis: Dumfume `51.2V 100Ah` LiFePO4, `3x` in parallel (`1S3P` ; manual allows up to `1S4P`).
- Per-battery charge voltage: `58.4V +/-0.2V` ; charge-limit/over-charge protection voltage also listed as `58.4V` .
- Per-battery current references: recommended charge `20A` ; max continuous charge `100A` ; recommended discharge `50A` ; max continuous discharge `200A` .
- Bank-level reference for current `3x` setup: recommended charge `60A` ; max continuous charge `300A` ; recommended discharge `150A` ; max continuous discharge `600A` .
- Protection thresholds: over-discharge protection `36.8V` , recovery `43.2V` ; discharge overcurrent protection `600A` ; short-circuit protection `1800A` .
- Temperature thresholds: low-temp charge protection approx `41F-50F` with recovery at `50F` ; high-temp protection listed as `157F` / `140F` .
- MultiPlus-II charger limit is `35A` , so full-output charging is below the bank's `60A` recommended-charge reference (`~11.7A` per battery in `3P`). For first household-outlet tests, keep AC input current at `10A` , then `12A` max on a normal `15A` source. Current charger-programming target is `56.8V` absorption/charge, `54.0V` float, `52.8V` storage if shown, `0.5h` max absorption, equalization off, and temperature compensation off.

48V power path



Alternator control path



Control logic

- `Upfitter #3 ON` : `WS500` brown wire is energized, regulator is allowed to run, alternator charging can occur.
- `Upfitter #3 OFF` : `WS500` is disabled through the brown wire, regulator field output collapses, and alternator charging stops.
- Use `Upfitter #3` as the manual alternator-charge shutdown control.
- Do not use the main `48V` disconnect as the first alternator shutdown method while the engine is running and the secondary alternator is charging.

Why `Upfitter #3`

- It is already a factory-switched, relay-driven `12V` control source.
- It keeps the WS500 control in the truck cab without adding a separate aftermarket switch.
- It is suitable for a low-current enable signal, but not for any high-current alternator or battery conductor.
- A local inline fuse is still required because the factory upfitter circuit fuse is much larger than the small-gauge WS500 control wire.

Fusing and wire baseline

ID	Function	Locked value	Wire basis
`F-01A/B/C`	Battery branch positive protection	`200A` Class T provisional	`2/0 AWG`
`F-04`	Alternator branch into Lynx Slot 3	`150A` MEGA (`58V/80V`)	`2/0 AWG`
`F-05`	Lynx Slot 4 -> Orion `48V` input	MEGA `40A`, body-marked `>=58VDC`; Victron `CIP138040020` `40A/80V` replacement fallback	Existing `6 AWG` direct to Orion; single input fuse
`F-06`	Retired standalone Orion input fuse position	Not installed	MIDI/FKS/DIN concepts superseded; do not stack with `F-05`
`F-12`	WS500 regulator power lead	`10A` baseline (`15A` if required by alternator case); voltage rating must cover the connected alternator/system positive voltage	harness lead

F-13	WS500 positive voltage-sense lead	3A; fuse/holder voltage class must cover the 48V bank maximum unless proven by WS500 harness documentation	harness lead
CERBO-PWR	Cerbo GX low-current power feed	1A-3A inline fuse/holder rated for the 48V bank maximum; system/load side of main disconnect preferred for bench shutdown	18 AWG red/black duplex acceptable
WS500 current-sense pair	Purple/grey current-sense high/low to shunt/current-sense point	No separate fuse position in current Wakespeed manual; twist pair if extended and route away from noise	harness lead
F-15	Upfitter #3 to WS500 brown ignition wire	3A inline ATO/ATC on 12V control circuit	16 AWG TXL/GXL control wire

Major conductors

- Battery branch and main `48V` trunk wiring: `2/0 AWG`.
- Parallel battery-current sharing target is similar **total loop resistance** per battery path, not equal positive-only length. The current bench layout may use short/medium/long positive leads balanced by long/medium/short negative leads; do not add unnecessary cable coils solely to make positive leads identical.
- Secondary alternator positive path (`ALT B+ -> APM-48 -> F-04 -> Lynx`): `2/0 AWG`, `~20 ft` one-way planning basis.
- Secondary alternator dedicated negative return (`ALT B- -> Lynx -`): `2/0 AWG`, `~20 ft` one-way planning basis.
- Orion `48V` feeder: existing `6 AWG` remains as the no-rework path from Lynx Slot 4 and is protected by `F-05 40A` MEGA (`>=58VDC`); it is electrically overkill, and flexible fine-strand `10 AWG` would be adequate if ever replaced for unrelated reasons. MPPT battery leads and Orion `12V` output remain `6 AWG`.
- Upfitter #3 control lead to WS500 brown wire: `16 AWG` TXL/GXL planning basis, `~6 ft` one-way assumed until measured.

APM-48 wiring intent

- Mount the `APM-48` at the alternator end of the `48V` branch, as close to the alternator output/ground points as practical.
- Treat the APM-48 as a parallel surge/load-dump clamp across the alternator, not a series device in the alternator charge-current path.
- Connect APM red to alternator `B+`.
- Connect APM black to alternator `B-`, or to the approved alternator ground/case point if the alternator is not isolated-ground.
- Do **not** place either APM connector under the main battery cable lugs; Balmar's quick-start sheet explicitly prohibits placing APM connectors under the battery cable lugs.

Normal operating sequence

1. Main `48V` disconnect closed.
2. `Upfitter #3` switched `ON`.
3. Engine running.
4. `WS500` enabled and regulating.
5. Alternator charges the house bank through `APM-48` and `F-04`.

Fault and shutdown sequence

If a battery pack trips or an alternator fault is suspected:

1. Switch `Upfitter #3` `OFF` to disable the `WS500`.
2. Wait for alternator charge current to collapse.
3. Open the main `48V` disconnect only after alternator charging is no longer active, if full house shutdown is required.

Reason:

- The regulator should be shut down first.
- The `APM-48` is a protection layer, not the primary shutdown method.

One-battery-trip behavior

- If one battery BMS disconnects but the other two remain online, the `48V` bus should stay up.
- The system does not immediately become a full-bank load dump event just because one battery disappears.
- The practical effect is that charge and discharge current redistribute to the remaining batteries.

- The real hazard is a cascade where the second and third battery also disconnect under active alternator charge.

What is still not fully closed

- Final Mechman kit fitment/content confirmation for the exact truck.
- Final `WS500` harness polarity confirmation (`PH` vs `NH`).
- Official vendor confirmation that the documented Dumfume battery/BMS behavior is acceptable with the `WS500` .
- Exact alternator negative/case isolation behavior in the installed Mechman kit.
- Whether future automatic fault-interlock logic should be added on the WS500 `Feature-In` or through an external relay.

Rule for future edits

- Update this file first for any `48V` architecture, alternator-control, or shutdown-strategy change.
- Keep the implementation files for wiring detail and fuse placement, not for high-level architecture decisions.